When Does Pooling Make Intersectional Fairness Audits Falsely Reassuring?

Bayesian audits of sparse race × sex × disability groups in ACSPublicCoverage

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Abstract

Aggregate fairness audits can hide disparities at intersections of protected attributes. Intersections are often too small for raw estimates, and hierarchical Bayesian pooling is the established remedy. We ask when such pooling makes an audit *falsely reassuring*, i.e. when it concludes that an equality-of-opportunity tolerance $\Delta\text{EOpp} \leq \epsilon$ holds when it does not. We use ACSPublicCoverage (California 2018; 138,554 people), a compact Bayesian-network predictor with exact inference, and a hypothetical outreach decision at 30% capacity. We compare raw, independent, exchangeable (empirical-Bayes and full-Bayes beta-binomial), attribute-structured (hierarchical logistic) and ϵ-contamination audits in synthetic, semi-synthetic and real experiments. Exchangeable pooling declares a 0.20 deficit in a 24-positive intersection "fair" in 98–100% of semi-synthetic worlds, and an upper-posterior-quantile rule does not fix this. Attribute-structured pooling masks the same deficit when it is an interaction. On real data, every one of 116 validation-time tolerance claims made by pooled criteria fails on an independent household-disjoint test split (test ΔEOpp 0.23–0.77). A contamination procedure calibrated on an invented design lowers false claims, but its acceptance of truly fair worlds falls from 0.83 to 0.32 on the real design. We trace this to groups with fewer than 10 positives and propose two fixes: *design-matched calibration* on the deployment population's own validation group sizes, and *held-out certification*. In a preregistered out-of-sample test on Texas 2018, the design-matched audit scores $J = \text{ACC} - \text{FTS} = 0.664$, against at most 0.385 for every established audit, at a false-certification rate of 0.168 (vs 0.615–0.661 for hierarchical pooling). Held-out certification removes all validation-time false certificates (10 of 10 same-data claims were false; held-out certification made none). Our falsifiable claim (H1), that pooled validation-time tolerance claims fail on test more than half the time, is supported out of sample on Texas 2018 (27 of 27 claims failed).

1 Introduction

Fairness audits usually report one gap per protected attribute. Disadvantage, however, often concentrates at intersections [3, 6]. Intersections of race, sex and disability are numerous and small, so raw estimates of group true-positive rates (TPRs) are noisy and the max−min gap is biased upward [10]. Hierarchical Bayesian models stabilise such estimates by partial pooling [1, 8, 11]. Pooling, however, pulls a small group toward the majority. If that group is the one that is actually disadvantaged, the audit may certify fairness that does not exist.

Research questions. **RQ1:** When does hierarchical pooling improve intersectional fairness assessment, and when does it mask a harmful small-group disparity? **RQ2:** How do pooling strength and

group size affect interval coverage? **RQ3:** Can posterior uncertainty (an upper posterior quantile) improve validation-time threshold selection? **RQ4:** Are conclusions stable across splits and populations?

Contributions. The literature already establishes hierarchical intersectional estimation, small-group unreliability, marginal-vs-intersectional gaps and contamination priors (Section 2). We therefore claim an *evaluation*, not a new estimator: (i) a controlled comparison of false tolerance-satisfaction rates (FTS) for six audit families, under minority thinning and three deficit structures, on both an invented and the real intersection design; (ii) evidence that pooled audit criteria used for validation-time rule selection make tolerance claims that fail on independent data; (iii) a diagnosis of why a simulation-calibrated robust-prior procedure fails to transfer between designs, and a fix, *design-matched calibration plus held-out certification*, tested out of sample with preregistered hypotheses (H2a–c); and (iv) an out-of-sample test, on a second state, of the falsifiable claim

> **H1.** *When the group-threshold rule's λ is selected on validation data and a pooled Bayesian audit criterion reports intersectional $\Delta EOpp \leq 0.10$, the frozen rule's empirical test $\Delta EOpp$ exceeds 0.10 in more than half of those cases.*

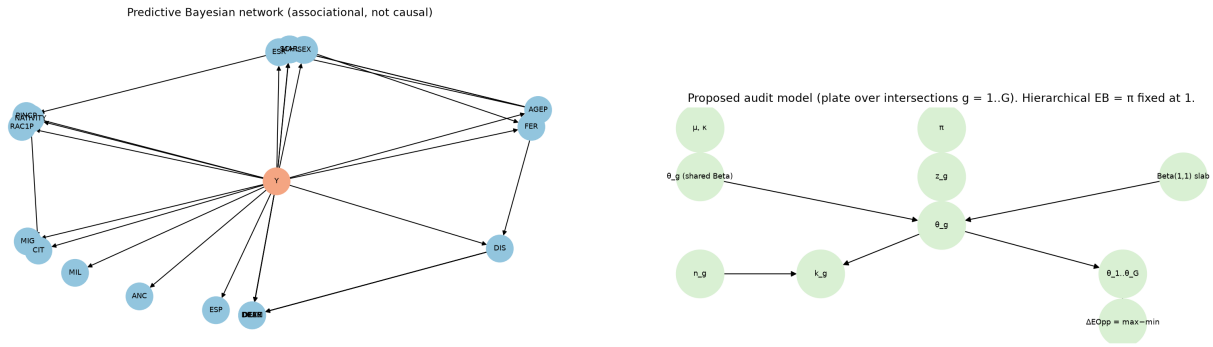
2 Related work

Intersectional definitions. Subgroup fairness over rich classes [3], multicalibration [4] and differential fairness [2] formalise intersectional fairness. Marginal fairness does not in general imply intersectional fairness [5]; a survey is given in [7]. *Small-group estimation.* Foulds et al. [1] introduce Bayesian models, including a hierarchical Bayesian logistic regression, for sparse intersections. Ji et al. [8] estimate fairness metrics with a hierarchical Bayesian calibration model. Herlihy et al. [11] use structured regression with confidence intervals for very small subgroups. Lum et al. [10] show that naive disparity estimates are biased. Himmelreich et al. [12] frame intersectionality as a problem of statistical uncertainty. Morina et al. [9] extend audit metrics to intersections. *Inference and generalisation.* Cherian and Candès [13] give simultaneous bootstrap bounds for subgroup audits. Cotter et al. [15] show that fairness constraints may not generalise and enforce them on validation data. Hardt et al. [14] define equality of opportunity and threshold post-processing. *Robust shrinkage.* ϵ-contaminated priors [19] and heavy-tailed hierarchical priors for outlying small areas [20] are classical. *Data and labels.* We use Folktables [16, 17]; label noise can overturn fairness conclusions [18]. The full review, with a comparison table of 19 papers, is in the companion document `literature_review.pdf`.

3 Data, decision and metrics

Data. ACS PUMS 2018 1-Year, California, through Folktables' `ACSPublicCoverage` task (AGEP $<$ 65, PINCP $\leq$ 30,000): 138,554 people in 85,182 households, 19 features identical to Folktables' output. The label is $Y = 1 - \mathbb{1}[\text{PUBCOV} = 1]$ (prevalence 0.631): $Y = 1$ means *no public coverage*, which is neither uninsurance nor eligibility. Protected attributes are RAC1P (nine codes, not collapsed), SEX and DIS, giving 36 intersections. Splits are household-disjoint 60/20/20 on SERIALNO with five seeds. Texas 2018 (98,928 people) is a replication population.

Decision. $D = 1$ selects a person for *hypothetical* information outreach, at a fixed capacity of 30%. The rule family is $t_g(\lambda) = (1 - \lambda)t_{\text{global}} + \lambda t_g^{\text{EOpp}} + c$, with c chosen so that exactly 30% of validation records are selected. $\lambda = 0$ is group-unaware; $\lambda = 1$ is ordinary opportunity-constrained threshold adjustment [14].



(a) Predictive BN

(b) Contamination audit model

Figure 1: Graphical models. (a) Associational predictor. (b) Audit model with a plate over intersections; hierarchical EB is the special case $\pi = 1$.

Metrics (preregistered). The primary fairness metric is intersectional $\Delta\text{EOpp} = \max_g t_g - \min_g t_g$ over groups with ≥ 30 positives, with tolerance $\epsilon = 0.10$ (robustness 0.05 and 0.15). The primary predictive metric is recall at capacity, with an acceptable utility loss of 0.02. ΔDP, ΔEOdds, R_{sel}, worst-group recall, AUROC, AP, log loss, Brier score and group calibration are diagnostics. In simulation we report FTS $= P(\text{declared fair} \mid \text{truly unfair})$, ACC $= P(\text{declared fair} \mid \text{truly fair})$, and the coverage and width of central 90% intervals.

4 Methods

4.1 Predictive Bayesian network and exact inference

The predictor is a discrete augmented naive-Bayes network: Y is a parent of all 18 features, plus edges such as PINCP|ESR and {DEAR, DEYE, DREM}|DIS (Figure 1a). CPTs are Dirichlet-smoothed ($\alpha = 1$) and $P(Y \mid x)$ is computed by variable elimination. Across 200 random evidence masks, our VE under two elimination orders, pgmpy VE, pgmpy clique-tree belief propagation and enumeration agree to 3.3×10^{-16}. The joint conditional independence SEX $\perp$ {AGEP, DIS} $\mid Y$ implied by d-separation holds in the fitted model. The graph is associational. Logistic regression and gradient-boosted trees are predictive baselines.

4.2 Audit models

For a frozen rule, let n_g be the positives in intersection g and k_g the selected positives, with $k_g \sim \text{Bin}(n_g, \theta_g)$ and θ_g the TPR. We compare: *raw* k_g/n_g; *independent* $\theta_g \sim \text{Beta}(1,1)$ (no pooling); *complete* pooling (one shared θ); *hier*, $\theta_g \sim \text{Beta}(a,b)$ with (a,b) fitted by beta-binomial marginal likelihood; *fullbayes*, $\text{Beta}(\mu\kappa, (1-\mu)\kappa)$ with $\mu \sim U(0,1)$, $p(\kappa) \propto (1+\kappa)^{-2}$, on a grid; *struct*, the attribute-structured hierarchical logistic audit $\text{logit}\,\theta_g = \beta_0 + \beta_{\text{race}} + \beta_{\text{sex}} + \beta_{\text{dis}} + u_g$, $u_g \sim N(0, \tau^2)$, with (β, τ) fitted by marginal likelihood using 201-node adaptive quadrature (a restricted version in the spirit of [1], not an exact reproduction); and *contamination*, $\theta_g \sim \pi\,\text{Beta}(a,b) + (1-\pi)\,\text{Beta}(1,1)$ with $\pi \leq \pi_{\max}$ [19]. Under the contamination prior the posterior is a two-component Beta mixture, so a group whose counts are implausible under the shared Beta keeps its own estimate. A *plug-in* rule declares fairness if the max−min of the posterior means is $\leq \epsilon$. An *upper-q* rule declares fairness if the q-quantile of the posterior of $\max_g \theta_g - \min_g \theta_g$ is $\leq \epsilon$, equivalently $P(\Delta > \epsilon \mid \text{data}) \leq 1 - q$.

4.3 Calibrating the tolerance procedure

Before any real-data selection, we chose the audit procedure on synthetic data with a preregistered rule: maximise Youden's $J = \text{ACC} - \text{FTS}$ over (estimator, rule) pairs. The synthetic design has 16 intersections with 4000 down to 8 positives, homogeneous TPRs 0.65 ± 0.015, and in the unfair worlds one 20-positive group 0.20 lower, with $R = 300$ replicates. The chosen procedure is then frozen.

4.4 Experiments

E1 compares single-attribute, two-attribute and intersectional audits. **E2** thins three disabled minority intersections in training and validation to 50, 25 and 10% while keeping the test split intact. **E3** compares raw, independent and pooled estimates on real counts. **E4** simulates homogeneous and exceptional worlds on the synthetic design and, *semi-synthetically*, on the real seed-0 validation design (32 intersections, 1 to 5,499 positives). The deficit sits at Black, female, disabled (`2|2|1`, 24 positives), thinned to 100, 50 and 25%, and takes three forms: an isolated *cell*; a pure *interaction*, in which the other 31 groups are shifted so that every race, sex and disability margin is exactly equal; and a *disability main effect*. **E5** selects λ on validation with each criterion (raw, independent, complete, hier, fullbayes, struct and contamination; plug-in and upper rules) and audits the frozen rule on test, with capacity matched. Ablations remove pooling, use complete pooling, and replace the upper quantile by the posterior mean. Robustness tests vary priors and $\pi_{\max}$, base TPR, ϵ, effect size and thinning, and add survey weights and Texas. **Statistics.** Evaluation uncertainty comes from 1,000 household-cluster bootstrap replicates of the seed-0 test split. Training uncertainty comes from 5 refits. The confirmatory family (Holm) is the calibrated procedure vs hier plug-in and vs raw plug-in, on test ΔEOpp. Everything else is exploratory.

5 Results

5.1 Intersectional audits reveal what single attributes hide (E1)

Predictive accuracy is moderate: BN AUROC 0.729 ± 0.004, LR 0.716 ± 0.004, GBT 0.768 ± 0.003 (Table 1). For the unconstrained BN rule, ΔEOpp is 0.162 by sex, 0.384 by disability, 0.468 for sex×disability and **0.746 intersectionally** (21 of 35 groups adequate; Table 2). Every disabled intersection has a TPR of at most 0.08. Disabled low-income adults mostly do have public coverage, so the minority of them without it is ranked low. Survey weighting confirms this (weighted TPR 0.011 and 0.031 for disabled men and women, against 0.280 and 0.448 without a disability). Texas replicates the pattern (0.094 by sex; 0.888 intersectionally).

5.2 Pooling on real counts (E3)

On real counts the exchangeable estimators move only groups with 1–7 positives (pull 0.05–0.14), and no group with ≥ 30 positives moves by more than 0.032 (Figure 2). The TPRs are bimodal by disability, so the shared Beta is diffuse. The struct model fits $\hat{\tau} = 0.01$ (its lower bound), so it becomes a main-effects model with an implausibly narrow gap interval, (0.750, 0.757). Every audit gives $P(\Delta\text{EOpp} > 0.10 \mid \text{data}) = 1.0$, so here pooling neither hides the disparity nor needs correcting.

5.3 Pooling masks a small exception; quantile rules do not fix it (E4, SYNTHETIC)

Table 3 and Figure 3a summarise the synthetic study. Empirical-Bayes pooling declares the unfair worlds fair in 100% (plug-in) and 98% (upper 0.9) of replicates, because shrinkage moves the ex-

Table 1: Test metrics of the unconstrained top-30% rule (mean ± sd over 5 household splits).

Metric	BN	LR	GBT
AUROC	0.729 ± 0.004	0.716 ± 0.004	0.768 ± 0.003
AP	0.803 ± 0.002	0.787 ± 0.002	0.836 ± 0.002
Recall@30%	0.388 ± 0.005	0.384 ± 0.006	0.409 ± 0.006
Prec.@30%	0.824 ± 0.004	0.809 ± 0.004	0.860 ± 0.002
Log loss	0.577 ± 0.004	0.584 ± 0.002	0.541 ± 0.003
Brier	0.196 ± 0.002	0.199 ± 0.001	0.182 ± 0.001
max group ECE	0.110 ± 0.010	0.069 ± 0.011	0.063 ± 0.013
ΔEOpp	0.719 ± 0.018	0.625 ± 0.049	0.585 ± 0.019
ΔDP	0.655 ± 0.016	0.548 ± 0.021	0.508 ± 0.013
worst recall	0.000 ± 0.000	0.000 ± 0.000	0.008 ± 0.011

Table 2: Single-attribute vs intersectional audit (seed-0 test split, unconstrained rule).

		BN			LR			GBT		
Grouping	adequate/all	ΔEOpp	ΔDP	worst rec.	ΔEOpp	ΔDP	worst rec.	ΔEOpp	ΔDP	worst rec.
SEX	2/2	0.162	0.128	0.291	0.080	0.060	0.338	0.095	0.077	0.346
DIS	2/2	0.384	0.324	0.021	0.396	0.334	0.010	0.354	0.307	0.067
RACE	8/9	0.597	0.548	0.079	0.561	0.456	0.106	0.360	0.313	0.169
SEX×DIS	4/4	0.468	0.386	0.009	0.436	0.361	0.004	0.427	0.352	0.034
RACE×SEX	14/18	0.678	0.614	0.059	0.611	0.517	0.099	0.429	0.410	0.155
RACE×DIS	13/18	0.697	0.634	0.000	0.686	0.554	0.000	0.544	0.464	0.000
RACE×SEX×DIS	21/35	0.746	0.683	0.000	0.711	0.580	0.000	0.600	0.519	0.000

ceptional group's posterior location, not just its interval width. Full Bayes with the upper 0.9 rule lowers FTS to 0.31, but accepts only 43% of fair worlds. Raw and independent audits never certify (ACC ≤ 0.007). No configuration achieves FTS < 0.10 with ACC > 0.5. The preregistered calibration selected the contamination prior with $\pi_{\max} = 0.8$ and the *plug-in* rule (FTS 0.503, ACC 0.830, $J = 0.327$). The best upper-quantile rule ($J = 0.323$) is within Monte Carlo error, so for RQ3 posterior quantiles did not improve the calibrated choice. Coverage of the 90% gap interval is 0.000 for hier and 0.038 for full Bayes, against 0.962 for the calibrated contamination prior. For RQ2, the struct audit under its own generative model covers only 0.665 at nominal 0.90 (0.53 for groups with < 30 positives). With the true hyperparameters the same sampler gives 0.905, so the shortfall comes from empirical-Bayes plug-in estimation.

5.4 On the real design: structure of the deficit matters, and calibration does not transfer (E4, SEMI-SYNTHETIC)

Table 4 gives the results on the real design. Hier declares the 24-positive cell deficit fair in 98–100% of worlds, and its posterior mean for `2|2|1` is 0.378 against a truth of 0.18. The attribute-structured audit also masks the deficit when it is an isolated cell or a pure interaction: its mean is 0.368–0.377, its interval coverage for `2|2|1` is 0.000 and its FTS is 0.23–0.37. When the deficit is a disability main effect shared by all 16 disabled groups, every estimator detects it. The calibrated procedure keeps FTS at 0.13–0.28 (higher with thinning), but its ACC is 0.32, against 0.83 in calibration. Many tiny groups give some group a chance to escape pooling in most fair worlds.

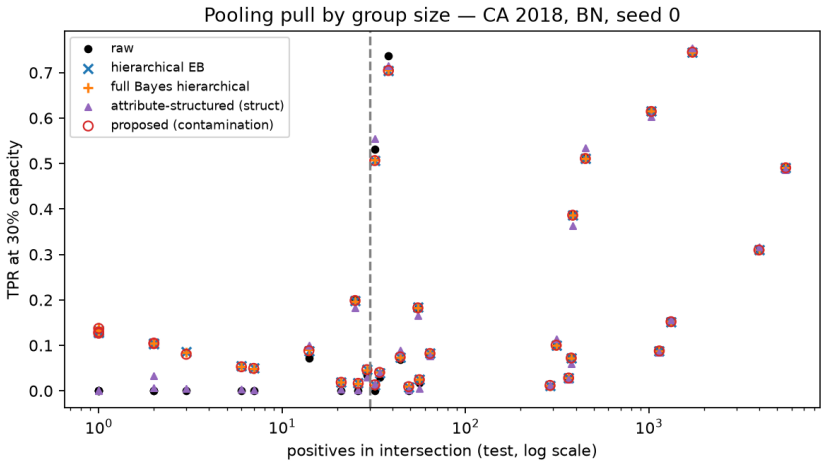


Figure 2: Raw and pooled TPR estimates by intersection size (BN, seed 0, test).

5.5 Validation-time tolerance claims fail on test (E2, E5)

Table 5 gives the selection results. Pooled criteria (hier, full Bayes, complete, the calibrated contamination rule, and struct in 2 of 5 splits) found a λ satisfying $\epsilon = 0.10$ on validation. Raw and independent criteria never did. **All 116 pooled claims across splits and thinning levels failed on the test split**, with test ΔEOpp from 0.227 to 0.770. Part of this is pooling and part is that thresholds fitted on small validation intersections do not transfer: the raw validation gap at $\lambda = 1$ is 0.48. Thinning to 10% raises the mean test gap to 0.376–0.410. Utility is within budget: recall is 0.388 at $\lambda = 0$ and 0.380 at $\lambda = 1$ (the cost is at most 0.013 in any split). Figure 4 shows the trade-off curves.

5.6 Confirmatory tests, refits and out-of-sample test of H1 (RQ4)

In the seed-0 bootstrap ($B = 1000$), the calibrated procedure and hier plug-in chose the same $\lambda = 0.9$ (difference exactly 0; Holm $p = 1$). Against raw plug-in the test ΔEOpp difference is -0.052, 95% CI $[-0.194, 0.051]$, Holm $p = 0.736$, at a recall cost of -0.006 $[-0.0089, -0.0035]$. Across the 5 refits the paired differences change sign (-0.016 ± 0.039 vs raw; $+0.009 \pm 0.035$ vs hier), so neither evaluation nor training uncertainty supports an improvement. An earlier seed-0 result (Holm $p = 0.04$) does not reproduce. Real-data conclusions are unchanged for $\epsilon \in \{0.05, 0.10, 0.15\}$ and for every prior and $\pi_{\max}$ tested. **Out-of-sample test of H1 on Texas 2018 (decision rule fixed in advance):** pooled criteria made 27 validation-time tolerance claims in 40 pooled selections, and **all 27 failed on the Texas test split** (test ΔEOpp 0.110–0.888). Raw and independent criteria made none. H1 is therefore supported on a population that played no part in forming it, although one claim failed only narrowly (0.110) and this is a single replication state.

Table 3: SYNTHETIC: FTS by thinning of the exceptional group, ACC, and 90% gap-interval coverage and width ($R = 300$).

	FTS (exceptional group kept)			ACC	90% gap interval	
Estimator · rule	100%	50%	25%	homog.	coverage	width
raw · plugin	0.003	0.000	0.000	0.007	–	–
independent · upper 0.9	0.000	0.000	0.000	0.000	0.264	0.382
complete · plugin	1.000	1.000	1.000	1.000	0.000	0.000
hier · plugin	1.000	1.000	1.000	1.000	0.000	0.018
hier · upper 0.9	0.980	0.987	0.987	0.993	0.000	0.018
fullbayes · plugin	0.983	1.000	0.993	1.000	0.038	0.114
fullbayes · upper 0.9	0.310	0.337	0.397	0.427	0.038	0.114
proposed $\pi \leq$0.8 · plugin (calibrated)	0.503	0.593	0.563	0.830	0.962	0.399
proposed $\pi \leq$0.8 · upper 0.5	0.450	0.567	0.577	0.773	0.962	0.399
proposed $\pi \leq$0.8 · upper 0.9	0.000	0.000	0.000	0.000	0.962	0.399
proposed $\pi \leq$0.95 · upper 0.9	0.297	0.300	0.317	0.527	0.589	0.273
proposed $\pi \leq$0.99 · upper 0.9	0.780	0.830	0.883	0.930	0.200	0.114

Table 4: SEMI-SYNTHETIC (real seed-0 validation design, invented TPRs; $R = 300$).

	ACC	FTS cell		FTS interaction		FTS dis.	$\theta_{2\|2\|1}$ (cell, keep 1)	
Procedure	keep 1	keep 1	keep .25	keep 1	keep .25	keep 1	90% cov.	post. mean
proposed (robust@0.8, plugin)	0.323	0.133	0.280	0.160	0.233	0.000	0.720	0.281
fullbayes · plugin	0.997	1.000	0.997	1.000	1.000	0.000	0.000	0.371
fullbayes · upper0.9	0.413	0.397	0.477	0.520	0.540	0.000	0.000	0.371
hier · plugin	1.000	1.000	0.997	1.000	1.000	0.000	0.000	0.378
hier · upper0.9	0.970	0.997	0.983	0.993	0.987	0.000	0.000	0.378
independent · plugin	0.000	0.000	0.000	0.000	0.000	0.000	0.900	0.201
independent · upper0.9	0.000	0.000	0.000	0.000	0.000	0.000	0.900	0.201
raw · plugin	0.000	0.000	0.000	0.000	0.000	0.000	–	–
struct · plugin	0.277	0.293	0.367	0.367	0.363	0.000	0.000	0.368
struct · upper0.9	0.217	0.233	0.327	0.320	0.313	0.000	0.000	0.368

Texas 2018, 5 household splits	value
pooled selections	40
validation-time claims ΔEOpp ≤ 0.10 (pooled)	27
claims failing on test (ΔEOpp > 0.10)	27
test ΔEOpp of the claims	0.110–0.888
claims by raw / independent criteria	0
H1 verdict	**supported**

5.7 Improving the audit: diagnosis, design-matched calibration, held-out certification (H2)

Diagnosis. In fair worlds on the CA design, 90.1% of the false alarms of the invented-design procedure come from a max- or min-group with fewer than 10 positives. Only 1.5% have both extreme groups

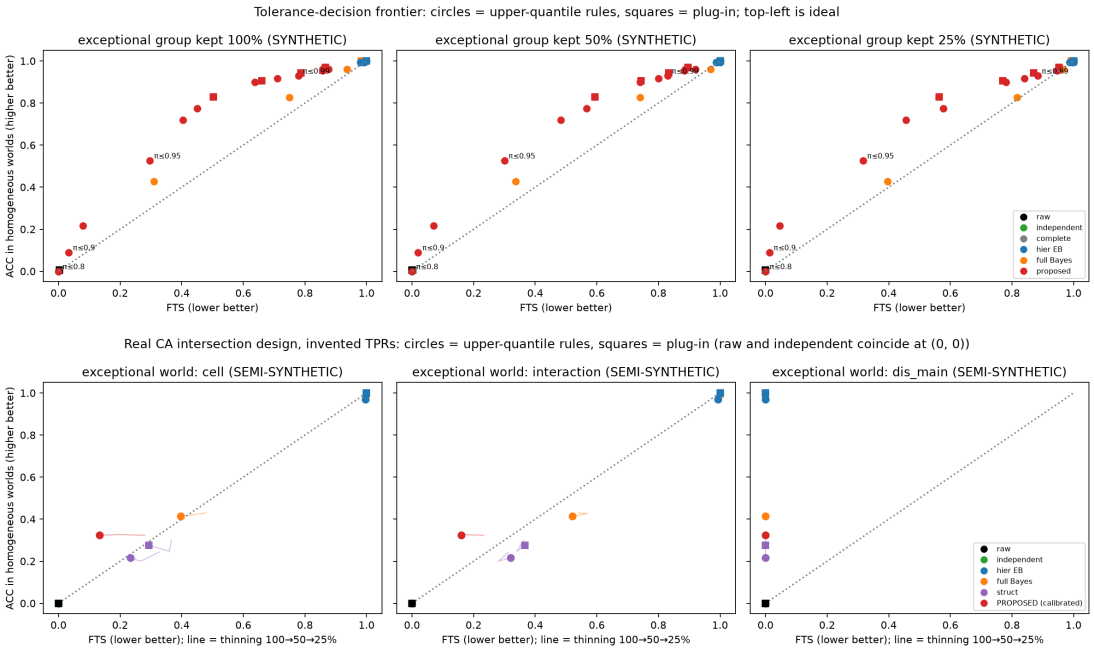


Figure 3: Tolerance-decision frontiers (top-left is ideal). Top: SYNTHETIC design. Bottom: SEMI-SYNTHETIC worlds on the real CA design, by deficit structure.

Table 5: Rule selection on validation, audited on test (intact data, 5 splits).

Selection criterion	feasible on val.	mean λ	test recall	test sel. rate	test ΔEOpp (mean $\pm$ sd)	worst recall	test $P(\Delta > \epsilon)$
raw · plug-in	0/5	0.80	0.385	0.299	0.348 $\pm$ 0.077	0.172	0.997
independent · upper	0/5	0.84	0.383	0.299	0.418 $\pm$ 0.094	0.191	0.996
complete · plug-in	5/5	0.26	0.390	0.298	0.641 $\pm$ 0.066	0.019	1.000
hier EB · plug-in	5/5	0.90	0.383	0.299	0.323 $\pm$ 0.077	0.210	0.965
hier EB · upper	5/5	1.00	0.380	0.299	0.367 $\pm$ 0.060	0.260	0.952
full Bayes · upper	5/5	1.00	0.380	0.299	0.367 $\pm$ 0.060	0.260	0.950
struct · plug-in	2/5	1.00	0.380	0.299	0.367 $\pm$ 0.060	0.260	0.955
struct · upper	2/5	1.00	0.380	0.299	0.367 $\pm$ 0.060	0.260	0.950
proposed · plug-in (calibrated)	5/5	0.96	0.381	0.299	0.332 $\pm$ 0.072	0.245	0.921
proposed · upper	0/5	1.00	0.380	0.299	0.367 $\pm$ 0.060	0.260	0.952

well supported. The tolerable amount of escape from pooling therefore depends on how many tiny groups a design has. **Fix.** (i) *Design-matched calibration:* the preregistered Youden rule chooses the slab (fixed Beta(1,1) or centred Beta($\mu c, (1-\mu)c$), $c = 2$), $\pi_{\max}$ and the rule on semi-synthetic worlds built from the deployment population's validation group sizes, with the deficit at a random group of 10–60 positives. (ii) *Held-out certification:* choose λ on one half of the validation households and certify on the other half. **Preregistered test.** Hypotheses H2a (J above every established audit by ≥ 0.10), H2b (J above the invented-design calibration by ≥ 0.10) and H2c (no false certificate on CA or TX test data) were fixed before running. They are evaluated on Texas, whose design played no part in developing the fix, with fresh seeds and the deficit at Texas's Black-female-disabled group. **Results.** All three are supported (Table 6, Figure 5, Table 7). On Texas the design-matched procedure (centred slab, $\pi \leq 0.95$, upper 0.5) has $J = 0.664$, against 0.385 for the best established audit and 0.189 for the invented-design calibration, and false certification 0.168, against 0.615–0.661 for full-Bayes and hierarchical pooling. On the California design, which the pilot used, it only ties the pooled audits on J (0.330 vs 0.333–0.334), with half their false-certification rate (0.348 vs 0.666–0.667). The gain there is in safety, not in J. The centred slab was chosen for Texas but not for California, so its value

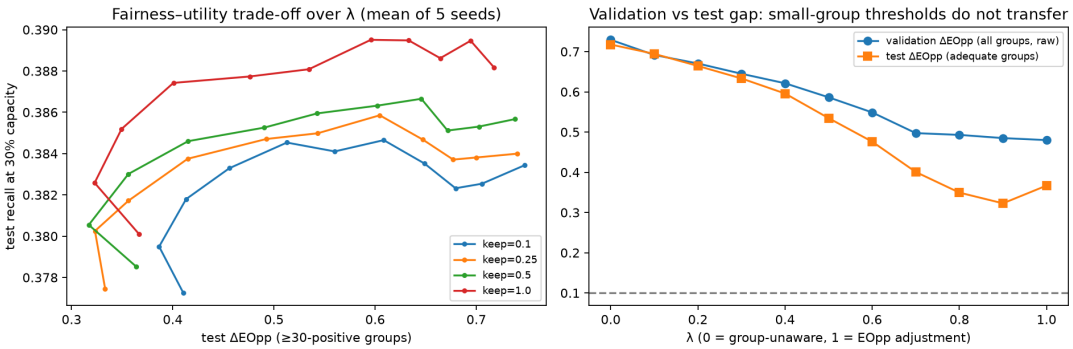


Figure 4: Fairness–utility trade-off over λ (left) and validation vs test gap (right), mean of 5 splits.

Table 6: Out-of-sample test on the Texas design (SEMI-SYNTHETIC, $R = 200$ per cell, keep 1 and 0.5). J = ACC− mean FTS.

	ACC	FTS			mean		90% gap
Procedure	(fair)	cell	interaction	dis. main	FTS	J	coverage
design-matched (centred-slab contam., $\pi \leq 0.95 \cdot$ upper 0.5)	0.832	0.255	0.250	0.000	0.168	**0.664**	0.615
fullbayes · plugin	1.000	0.920	0.925	0.000	0.615	**0.385**	0.174
hier · upper 0.9	0.985	0.930	0.945	0.000	0.625	**0.360**	0.013
hier · plugin	1.000	0.990	0.992	0.000	0.661	**0.339**	0.013
fullbayes · upper 0.9	0.290	0.110	0.145	0.000	0.085	**0.205**	0.174
§9 procedure (contam., $\pi \leq 0.8 \cdot$ plugin)	0.218	0.048	0.038	0.000	0.028	**0.189**	0.506
struct · plugin	0.108	0.125	0.095	0.000	0.073	**0.034**	0.037
struct · upper 0.9	0.090	0.105	0.078	0.000	0.061	**0.029**	0.037
independent · upper 0.9	0.000	0.000	0.000	0.000	0.000	**0.000**	0.000
raw · plugin	0.000	0.000	0.000	0.000	0.000	**0.000**	–

is design-dependent. Held-out certification made no false claims in either state, against 10 of 10 false claims with same-data certification, at unchanged recall. Because no rule in this family is truly fair, the real-data result shows that held-out certification avoids false certificates, not that it would certify a fair rule; the semi-synthetic ACC of 0.832 speaks to the latter.

6 Discussion

RQ1. Pooling masks a disparity when the deficit is confined to a small intersection and is not carried by any attribute the model pools through. Exchangeable pooling then hides it almost always, and attribute-structured pooling hides it when it is an interaction. Pooling is harmless when the disparity follows a main effect or involves large groups, as on real CA data, where the disparity is structural (disability) and huge. The most consequential real-data failure is selection-time: pooled criteria certified tolerances on validation that did not hold on test. **RQ2.** Interval coverage degrades with stronger pooling and smaller groups, and empirical-Bayes plug-in intervals under-cover even under a correct model. Bayesian credible intervals must not be read as frequentist guarantees. **RQ3.** Upper posterior quantiles did not rescue exchangeable pooling, and the invented-design calibration preferred a posterior-mean rule. Under design-matched calibration, however, upper-quantile rules were chosen in both states, and that procedure is the best audit out of sample, so posterior uncertainty helps once calibration matches the design. **RQ4.** No confirmatory difference is robust across bootstrap or refits. The qualitative real-data pattern replicates in Texas.

Limitations. The struct audit is a restricted version of [1], not an exact reproduction. The

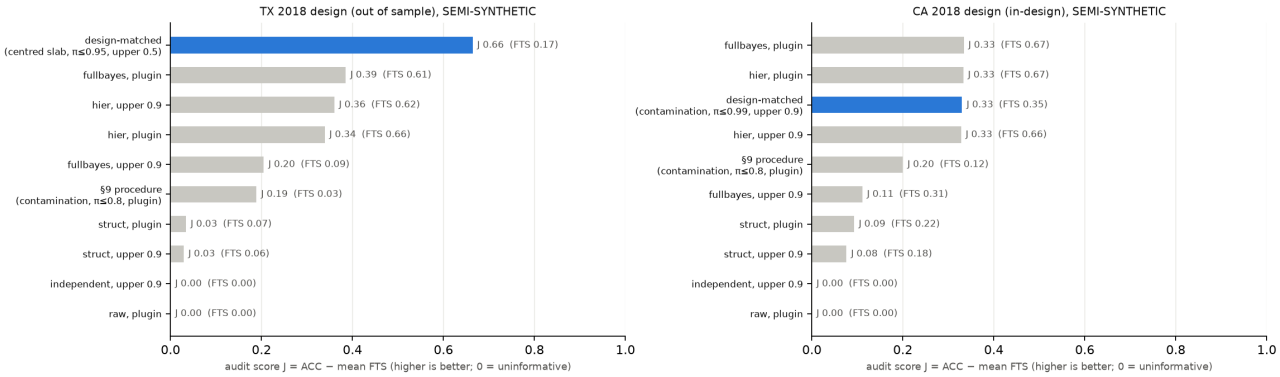


Figure 5: Audit score J per procedure. Left: Texas design (out of sample). Right: California design (in-design).

Table 7: Real data: validation-time tolerance claims of the design-matched procedure (5 household splits per state).

State	certification	claims ($\leq \epsilon$)	false claims	mean test ΔEOpp	mean test recall
CA	same-data	5/5	5	0.367	0.380
CA	held-out	0/5	0	0.351	0.379
TX	same-data	5/5	5	0.238	0.350
TX	held-out	0/5	0	0.267	0.350

synthetic calibration design is not representative of the real design. Binomial audit likelihoods ignore household clustering, and the label may contain group-dependent noise [18]. We did not reproduce simultaneous frequentist bounds [13]. Simulated effects are not facts about California. **Ethics.** The rule almost never reaches disabled people without public coverage. Pooling can erase a small group's experience. ϵ and the capacity were chosen by the analysts, not by the affected communities. No configuration here is "fair"; absence of public coverage must not be read as uninsurance or eligibility.

7 Conclusion

Bayesian pooling makes intersectional audits more stable, but it can make them falsely reassuring exactly where audits matter: small intersections whose disadvantage is not explained by any single attribute, and validation-time rule selection. Auditors should report raw counts, the pooling pull and $P(\Delta > \epsilon)$ together. They should never certify a tolerance from pooled estimates when intersections have fewer than about 30 positives. Our results support two concrete practices: calibrate the audit procedure on the design where it will be used (J 0.189 $\rightarrow$ 0.664 out of sample), and certify on data not used to choose the rule (zero false certificates, against 10 of 10 with same-data certification).

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A Reproducibility

Code and environment. All analysis is in the executed notebook `Problem8_Intersectional_Fairness.ipynb`, which runs top to bottom in about 30 minutes on one CPU core. Environment: Python 3.12 with the packages pinned in `requirements-lock.txt` (numpy 2.5.4, scipy 1.18.1, pandas 3.1.0, scikit-learn 1.9.1, pgmpy 1.1.2, folktables 0.0.12). **Data.** The notebook downloads ACS PUMS 2018 for CA and TX through Folktables. The data are not redistributed. **Determinism.** All seeds are fixed (splits 0–4, bootstrap 7, simulations fixed per section); two consecutive full runs produced identical printed outputs. **Commands.**

```
uv venv --python 3.12 .venv
uv pip install --python .venv/bin/python -r requirements.txt
.venv/bin/python -m ipykernel install --sys-prefix --name p8
.venv/bin/jupyter nbconvert --to notebook --execute --inplace \
    Problem8_Intersectional_Fairness.ipynb --ExecutePreprocessor.timeout=7200
```

Outputs. `results/*.csv` (all tables) and `figures/*.png`. The tables in this paper are generated from those CSVs by `writeup/make_tables.py`.

B Additional tables

Table 8: Intersectional audit table for the BN rule (unconstrained top-30%, seed-0 test split). n = people, n^+ = positives (no public coverage), k = selected positives, TPR = k/n^+. Posterior means under each audit model; pull = hier − raw. The contamination column uses $\pi_{\max} = 0.95$ because this table is computed (notebook §7) before the §9 calibration fixes $\pi_{\max} = 0.8$. $*$ = exploratory (< 30 positives or < 30 negatives).

Code	Race (RAC1P)	Sex	Disab.	n	n^+	k	raw TPR	indep.	hier	struct	contam. $\pi \leq 0.95$	pull
7|2|1	NHPI	F	yes	4	0	0	–	–	–	–	–	–*
4|2|2	Alaska Native	F	no	1	1	0	0.000	0.324	0.129	0.000	0.137	0.129*
4|2|1	Alaska Native	F	yes	2	1	0	0.000	0.330	0.130	0.000	0.130	0.130*
4|1|1	Alaska Native	M	yes	1	1	0	0.000	0.337	0.132	0.000	0.127	0.132*
5|2|1	AIAN, tribe spec./n.s.	F	yes	9	2	0	0.000	0.248	0.105	0.006	0.104	0.105*
7|1|1	NHPI	M	yes	8	2	0	0.000	0.252	0.103	0.033	0.105	0.103*
3|2|1	American Indian	F	yes	16	3	0	0.000	0.197	0.085	0.005	0.080	0.085*
5|1|1	AIAN, tribe spec./n.s.	M	yes	14	6	0	0.000	0.124	0.055	0.003	0.053	0.055*
3|1|1	American Indian	M	yes	20	7	0	0.000	0.110	0.050	0.003	0.049	0.050*
5|1|2	AIAN, tribe spec./n.s.	M	no	24	14	1	0.071	0.127	0.088	0.100	0.088	0.017*
2|2|1	Black	F	yes	142	21	0	0.000	0.044	0.019	0.003	0.019	0.019*
5|2|2	AIAN, tribe spec./n.s.	F	no	37	25	5	0.200	0.221	0.199	0.183	0.199	−0.001*
2|1|1	Black	M	yes	187	26	0	0.000	0.036	0.017	0.002	0.017	0.017*
9|2|1	Two or more	F	yes	77	29	1	0.034	0.064	0.047	0.030	0.047	0.013*
9|1|1	Two or more	M	yes	97	32	0	0.000	0.030	0.013	0.015	0.013	0.013
7|1|2	NHPI	M	no	48	32	17	0.531	0.528	0.506	0.554	0.506	−0.025*
6|1|1	Asian	M	yes	125	34	1	0.029	0.056	0.040	0.040	0.041	0.011
7|2|2	NHPI	F	no	57	38	28	0.737	0.725	0.704	0.716	0.705	−0.032*
3|1|2	American Indian	M	no	84	44	3	0.068	0.088	0.074	0.089	0.075	0.006
8|1|1	Some other race	M	yes	209	49	0	0.000	0.019	0.009	0.002	0.009	0.009
3|2|2	American Indian	F	no	89	55	10	0.182	0.192	0.183	0.165	0.183	0.002
8|2|1	Some other race	F	yes	197	56	1	0.018	0.035	0.025	0.005	0.024	0.007
6|2|1	Asian	F	yes	159	64	5	0.078	0.091	0.083	0.077	0.082	0.005
1|1|1	White	M	yes	965	289	3	0.010	0.014	0.012	0.012	0.012	0.002
2|2|2	Black	F	no	541	312	31	0.099	0.102	0.100	0.114	0.100	0.001
1|2|1	White	F	yes	1046	365	10	0.027	0.030	0.028	0.025	0.028	0.001
2|1|2	Black	M	no	655	377	27	0.072	0.074	0.073	0.060	0.072	0.001
9|1|2	Two or more	M	no	547	384	149	0.388	0.389	0.387	0.362	0.386	−0.001
9|2|2	Two or more	F	no	643	450	231	0.513	0.513	0.512	0.534	0.511	−0.001
6|1|2	Asian	M	no	1488	1030	634	0.616	0.615	0.615	0.602	0.614	0.000
8|1|2	Some other race	M	no	1824	1139	100	0.088	0.088	0.088	0.084	0.088	0.000
8|2|2	Some other race	F	no	2273	1320	201	0.152	0.153	0.152	0.156	0.152	0.000
6|2|2	Asian	F	no	2283	1727	1288	0.746	0.745	0.745	0.753	0.745	−0.001

Code	Race (RAC1P)	Sex	Disab.	n	n^+	k	raw TPR	indep.	hier	struct	contam. $\pi \le 0.95$	pull
`1|1|2`	White	M	no	5805	3980	1233	0.310	0.310	0.310	0.317	0.310	0.000
`1|2|2`	White	F	no	7976	5570	2734	0.491	0.491	0.491	0.486	0.491	0.000

Paired bootstrap, seed 0 ($B = 1000$).

Comparison	family	metric	estimate	95% CI	p_{boot}	p_{Holm}
calibrated − hier plug-in	confirmatory	ΔEOpp	+0.0000	[+0.0000, +0.0000]	1.000	1.000
calibrated − hier plug-in	confirmatory	recall	+0.0000	[+0.0000, +0.0000]	1.000	–
calibrated − raw plug-in	confirmatory	ΔEOpp	−0.0517	[−0.1944, +0.0505]	0.368	0.736
calibrated − raw plug-in	confirmatory	recall	−0.0061	[−0.0089, −0.0035]	0.001	–
proposed upper − hier plug-in	exploratory	ΔEOpp	−0.0714	[−0.1755, +0.0000]	0.074	–
proposed upper − hier plug-in	exploratory	recall	−0.0026	[−0.0048, −0.0005]	0.014	–
proposed upper − raw plug-in	exploratory	ΔEOpp	−0.1231	[−0.3073, +0.0091]	0.076	–
proposed upper − raw plug-in	exploratory	recall	−0.0086	[−0.0124, −0.0053]	0.001	–
struct upper − hier plug-in	exploratory	ΔEOpp	−0.0714	[−0.1755, +0.0000]	0.074	–
struct upper − hier plug-in	exploratory	recall	−0.0026	[−0.0048, −0.0005]	0.014	–
struct upper − raw plug-in	exploratory	ΔEOpp	−0.1231	[−0.3073, +0.0091]	0.076	–
struct upper − raw plug-in	exploratory	recall	−0.0086	[−0.0124, −0.0053]	0.001	–
struct plug-in − hier plug-in	exploratory	ΔEOpp	−0.0714	[−0.1755, +0.0000]	0.074	–
struct plug-in − hier plug-in	exploratory	recall	−0.0026	[−0.0048, −0.0005]	0.014	–
struct plug-in − raw plug-in	exploratory	ΔEOpp	−0.1231	[−0.3073, +0.0091]	0.076	–
struct plug-in − raw plug-in	exploratory	recall	−0.0086	[−0.0124, −0.0053]	0.001	–